5. Review of 1225

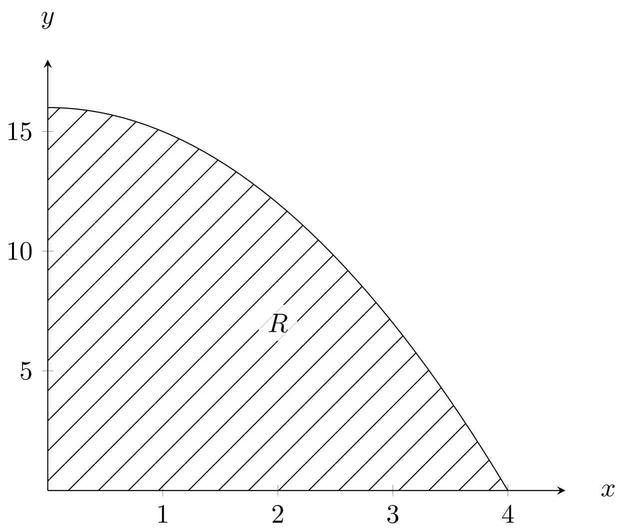
5.2. The Definite Integral

In this section, we will primarily review Riemann sums. This is because Riemann sums are used to derive many of the techniques used in the upcoming sections.

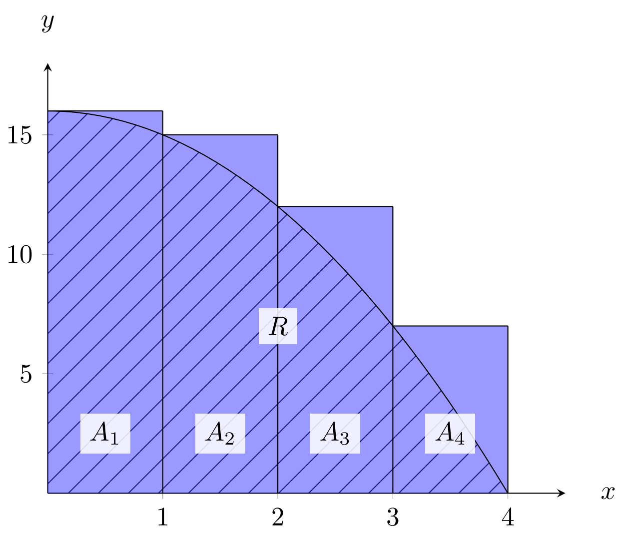
5.2 Learning Objectives:
- I can approximate the (signed/unsigned) area under a curve using different types of Reimann sums.
- I can use more rectangles/sub-intervals to get a more accurate approximation.
- I know the definition of a definite integral in terms of the limit or Reimann sums.
- I can rewrite a limit of Reimann sums as a definite integral, and back.

5.2.1. Motivation

Suppose we wanted to calculate the area of a shaded region R that lies above the x-axis, below the graph of $f(x) = -x^2 + 16$, and between the vertical lines $x = 0$ and $x = 4$.



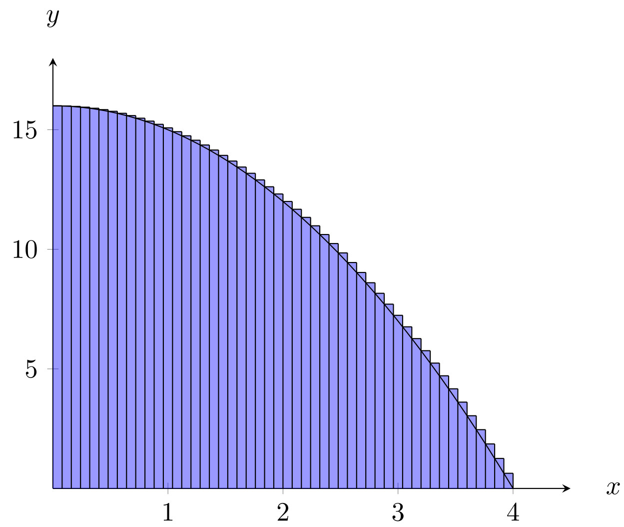
How might we estimate this area with rectangles?



The area can be estimated by summing the areas of the rectangles $R \approx A_1 + A_2 + A_3 + A_4$.

Remark

- We also aligned the left side of the rectangles to the curve, but we could have used any point between the left and right sides of the rectangles to get the height of the rectangles.
  - For example, right end point and midpoint are both popular choices.
- We used rectangles insteads of another shape, because the area of a rectangle is easy to find. We only need width and height, and the height is just a single function evaluation.
  - For example, the area of A_1 is $A_1 = (1)(f(0)) = (1)(16) = 16$.
  - Another shape you may encounter (outside this class) is the trapezoid, using the left AND right end points of the domain to get the 2 required heights for a trapezoid.
- We used 4 rectangles, but any number of rectangles can be used.
  - Using more (smaller) rectangles improves the accuracy of our estimated area. TODO label figure and reference it here



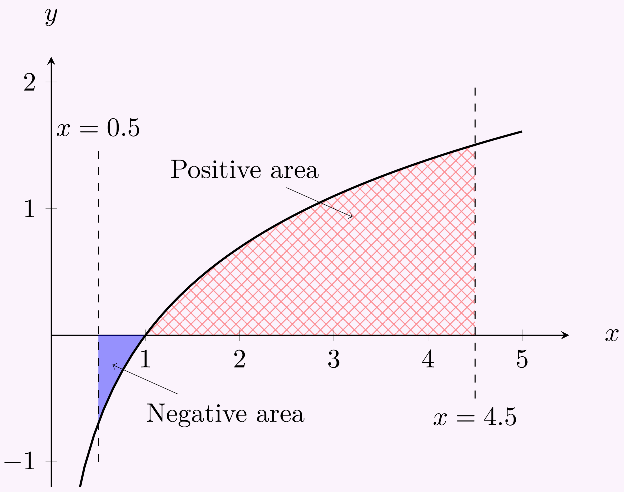
- In the above cases, the approximate areas have been overestimates of the actual area. In general, it is not possible to tell when we have an over or under estimate by looking at a picture. However, there are some cases we can always tell.
  - If the function is decreasing, then the right Riemann sums will underestimate and the left Riemann sums will overestimate.
  - If the function is increasing, then the right Riemann sums will overestimate and the left Riemann sums will underestimate.
  - If the function changes between increasing and decreasing, then we cannot guarantee one way without more information.

Exercise 5.2.1

1. Draw a few pictures of different increasing and decreasing functions and some left/right Reimann sums to get a feel for why they are always over/underestimates.
2. Does concavity have anything to do with the approximations being over or under estimates of the area under the curve?

Example 5.2.2

Suppse you want to find the *signed* area of the region R that lies between the x-axis, the graph of $f(x) = \ln(x)$, and between the vertical lines $x = 0.5$ and $x = 4.5$. The exact signed area would look something like this.

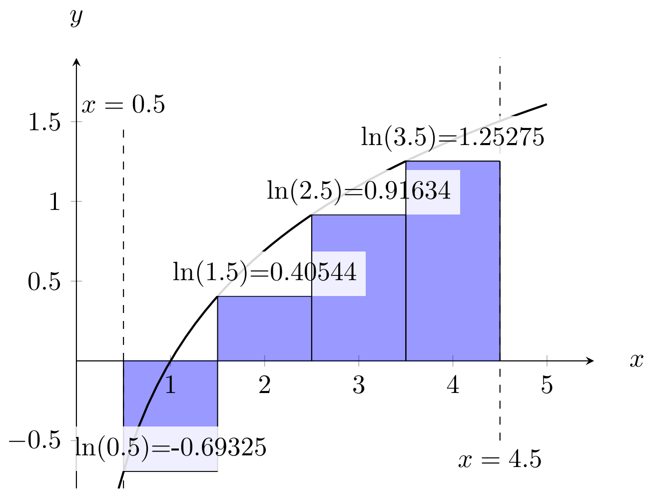


Approximate the signed area (as described above) to two decimal places with 4 rectangles using:

1. Left-hand Riemann sum
2. Right-hand Reimann sum
3. Mid-point Riemann sum

Solution

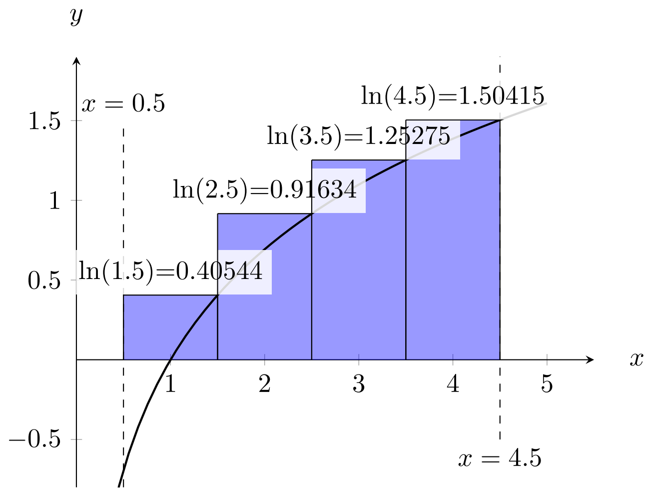
1. The left hand Reimann sum looks like the below image. Note, the area of the first rectangle is negative, because it is below the x-axis.



Thus,

$$\begin{aligned} L_4 &= A_1 + A_2 + A_3 + A_4 \\ &= (1)(\ln(0.5)) + (1)(\ln(1.5)) + (1)(\ln(2.5)) + (1)(\ln(3.5)) \\ &= \underbrace{1}_{\Delta x}(\ln(0.5) + \ln(1.5) + \ln(2.5) + \ln(3.5)) \\ &\approx 1.88 \end{aligned}$$

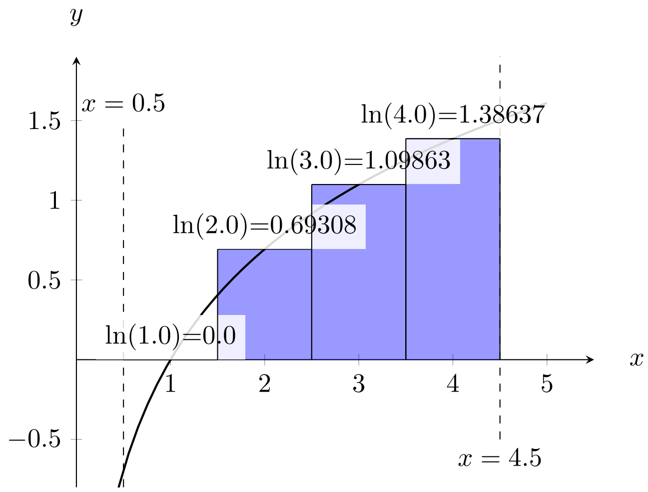
2. The right hand Reimann sum looks like the below image. Note, the area of the first rectangle is **no longer negative**, because it is now above the x-axis.



Thus,

$$\begin{aligned} R_4 &= (1)(\ln(1.5)) + (1)(\ln(2.5)) + (1)(\ln(3.5)) + (1)(\ln(4.5) \\ &= \underbrace{1}_{\Delta x}(\ln(1.5) + \ln(2.5) + \ln(3.5) + \ln(4.5)) \\ &\approx 4.08 \end{aligned}$$

3. The mid-point Reimann sum looks like the below image. Note, the first rectangle has height 0 since $\ln(1) = 0$.



Thus,

$$\begin{aligned} M_4 &= (1)(\ln(1)) + (1)(\ln(2)) + (1)(\ln(3)) + (1)(\ln(4) \\ &\approx 3.18 \end{aligned}$$

Remark

- The actual signed area is 3.11 rounded to 2 decimal places.
- This does not give the (unsigned) area (ignoring the fact that some parts are negative).
- We can find the unsigned area (counting all area as positive) by using $-f(x)$ for the rectangle height for the points where $f(x)$ is negative.
- As before, using more rectangles can give us a more accurate estimate of the areas (signed and unsigned). *Let's push this idea to the limit...*

Definition 5.2.3

Let f be a function defined on the interval $[a, b]$. We can divide the interval $[a, b]$ into n sub-intervals of equal width $\Delta x = \frac{b-a}{n}$. Let $x_0 = a, x_1, x_2, ..., x_{n-1}, x_n = b$ be the endpoints of these sub-intervals. Let $x_1^*, x_2^*, ..., x_n^*$ be any sample points in these sub-intervals such that x_i^* is in the sub-interval $[x_{i-1}, x_i]$. Then the **definite integral of f from a to b** is defined as

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

♣

Remark

- The definite integral $\int_a^b f(x)\,\mathrm{d}x$ represents the **exact *signed* area** under the curve $y = f(x)$ between a and b.
- The integral symbol is a stretched "S" shape with the **lower** and **upper bound** above and below the symbol respectively ("S" is for sum). The $\mathrm{d}x$ at the end closes the integral but also indicates that we are integrating with respect to x.
  - Eventually we will have other variables or multiple variables we are integrating over, so this is important for clarity.
  - The variable we are integrating over says where the "base" of the rectangles will be.
- As we saw in the previous examples, as we use more rectangles to approximate the area, we get closer to the exact signed area we are looking for. In the definition of the definite integral, n is the number of rectangles, and when we take the limit as n goes to infinity we are sub dividing into more and more rectangles to approximate the area. Eventually the rectangles approach "infinitesimally thin rectangles," that add to the exact area we are looking for.
- We have been talking about left-hand, right-hand, and mid-point Reimann sums, but we could also use a point 10% of the way into the sub-interval, or 30% into the sub-interval, or anywhere inside each sub-interval. We write this using x_i^* to indicate a generic choice of left/right/mid-point sample point in the i-th rectangle. The i here is the most important part, since it means this point depends on which rectangle we are in. The star is also important, since without it the definition would imply it dentoes the end-points of the sub-intervals.
  - You could theoretically use a different point for each rectangle, but we won't. Keep it simple and consistent. Use all left end points or all right end points etc depending on your choice for a problem.
- You should become comfortable translating between a definite integral and a limit of Reimann sums and back again. An example is below, come to office hours to discuss how starting at a Reimann sum does not lead to 1 correct definite integral, but many possible ones.

Example 5.2.4

Write

$$\int_1^3 \sqrt[3]{x^2+2}\,\mathrm{d}x$$

as a limit of Reimann sums.

Solution

We are not told to use a left or right Reimann sum specifically, so we can choose either. We will do a right-hand Reimann sum here (Practice on your own: the left one). The **bounds** on the integral give us $\boldsymbol{a=1}$ **and** $\boldsymbol{b=3}$**.** We can also clearly see the function is

$$f(x) = \sqrt[3]{x^2+2}.$$

The next thing to do is to write the width of the rectangles in terms of n:

$$\Delta x = \frac{b-a}{n} = \frac{2}{n}.$$

Lastly, we need a way to represent the right-end points of the rectangles in terms of i and n, and we will try to find a pattern for this. The **left** side of the first rectangle is at 1, and the next is Δx to the right of it, the next is another Δx to the right, and so on. The **right** sides of the rectangles skip 1 and start at $1+\Delta x$, then add another Δx every step. Repeated addition is multiplication so

$$x_i^* = 1 + \frac{2i}{n}.$$

All that's left is to put it altogether

$$\int_1^3 \sqrt[3]{x^2+2}\,\mathrm{d}x = \lim_{n\to\infty}\sum_{i=1}^{n} \sqrt[3]{\left(1+\frac{2i}{n}\right)^2+2}\left(\frac{2}{n}\right).$$

Example 5.2.5

Interpret

$$\lim_{n\to\infty}\sum_{i=1}^{n} \frac{2\left(3+\frac{3i}{n}\right)}{1+\left(3+\frac{3i}{n}\right)^2}\left(\frac{3}{n}\right)$$

as a definite integral.

Solution

For Δx we are looking for a number over n to show up, sometimes multiplied by i. We see a $\frac{3}{n}$ in a few places that fits this bill, so we will say $\Delta x = \frac{3}{n}$. Then, we try to identify x_i^*, the sample points. These have $\Delta x i$ in them, we can pick just the $\frac{3i}{n}$ or the $3+\frac{3i}{n}$. Either way we can get **a** correct answer. Let's pick $x_i^* = 3+\frac{3i}{n}$, then **in this case** we must have

$$f(x_i^*) = \frac{2x_i^*}{1+(x_i^*)^2}$$

(by replacing the $3 + \frac{3i}{n}$ with x_i^*). Lastly, all the sample points x_i^* "start" at 3 plus something: so $a = 3$. Then since $\Delta x = \frac{3}{n}$ we know the width of the interval is 3 (see numerator), so $b = 6$. Putting that all together

$$\lim_{n\to\infty} \sum_{i=1}^{n} \frac{2(3+\frac{3i}{n})}{1+(3+\frac{3i}{n})^2}\left(\frac{3}{n}\right) = \int_3^6 \frac{2x}{1+x^2}\,\mathrm{d}x.$$

Exercise 5.2.6

Redo the above problem but using $\Delta x = \frac{3i}{n}$. Notice how this changes the function and the bounds of the integral. To check your answer plug in both integrals into an integral calculator (https://www.integral-calculator.com/ or a TI 84 or more should do it). You know you are correct if you get the same number for both!

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Property 5.2.7 (Properties of the Definite Integral)

Let f and g be continuous functions. Let c be a constant.

- $\int_b^a f(x)\,\mathrm{d}x = -\int_a^b f(x)\,\mathrm{d}x$
- $\int_a^a f(x)\,\mathrm{d}x = 0$
- $\int_a^b c\,\mathrm{d}x = c(b-a)$
- $\int_a^b f(x) \pm g(x)\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x \pm \int_a^b g(x)\,\mathrm{d}x$
- $\int_a^b cf(x)\,\mathrm{d}x = c\int_a^b f(x)\,\mathrm{d}x$
- $\int_a^b f(x)\,\mathrm{d}x + \int_b^c f(x)\,\mathrm{d}x = \int_a^c f(x)\,\mathrm{d}x$
- If $f(x) \geq 0$ for all $a \leq x \leq b$, then $\int_a^b f(x)\,\mathrm{d}x \geq 0$
- If $f(x) \geq g(x)$ for all $a \leq x \leq b$, then $\int_a^b f(x)\,\mathrm{d}x \geq \int_a^b g(x)\,\mathrm{d}x$
- If $m \leq f(x) \leq M$ for all $a \leq x \leq b$, then $m(b-a) \leq \int_a^b f(x)\,\mathrm{d}x \leq M(b-a)$

♠

Exercise 5.2.8

1. Rewrite in your own words what each property above is saying/does. You may have to think about some of these a while, and you may discuss in office hours.
2. Justify/Prove the third property using a sketch and a few words. ($\int_a^b c \, \mathrm{d}x = c(b-a)$)
3. Justify to yourself why the penultimate property is true. (If $f(x) \geq g(x)$ for all $a \leq x \leq b$, then $\int_a^b f(x) \, \mathrm{d}x \geq \int_a^b g(x) \, \mathrm{d}x$)
4. Using the above properties prove the last property. (If $m \leq f(x) \leq M$ for all $a \leq x \leq b$, then $m(b-a) \leq \int_a^b f(x) \, \mathrm{d}x \leq M(b-a)$) This should be done in 2 steps since there are 2 inequalities here. *This is not trivial, so spend a bit of time thinking and working on this. The process of doing this yourself should deepen your understanding of what is going on here, if you get stuck, bring your work to office hours to discuss.*

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5.2 Section Summary:
- We discussed area beneath a curve, and discussed Reimann sums as ways to approximate the area.
- We discussed the difference between signed and unsigned area.
- We took the limit of Reimann sums to get the exact *signed* area, this became our definition of the definite integral.
- We practiced translating between Reimann sums and definite integrals.
- We looked at several properties of integrals.

5.5. The Substitution Rule

Recall that we linked the definite integral and area under a curve to the concept of the antiderivative. We did this using the fundamental theorem of Calculus.

5.5 Learning Objectives:
- I can use a substitution to undo the chain rule when antidifferentiating.
- I can change the bounds of a definite integral when I make a substitution.
- I can try a few different things for my substitution before giving up.

Theorem 5.5.1 (The Fundamental Theorem of Calculus, Pt I)

Let f be a function defined on the interval $[a, b]$. Let F be the function defined, for all x in $[a, b]$, by $F(x) = \int_a^x f(t) \, \mathrm{d}t$. Then F is continuous on $[a, b]$ and differentiable on (a, b) and $F'(x) = f(x)$ for all x in (a, b) so F is an antiderivative of f.

♡

Corollary 5.5.1.1 (The Fundamental Theorem of Calculus, Pt II)

If f is a continuous function defined on $[a, b]$ and F is an antiderivative of f in $[a, b]$ then

$$\int_a^b f(t) \, \mathrm{d}t = F(b) - F(a).$$

♡

This allows us to start evaluating definite integrals without sorrying about limits of sums. Instead, to find the area under a curve, we just need an antiderivative of it.

Exercise 5.5.2

Find an antiderivative for the following functions. Check your work by taking the derivative of your answer.
1. $f(x) = x^3 - 2x + 1$
2. $g(x) = (x + 1)^{10}$

This is fine for simple functions that we can recognize that we are looking at the derivative of something. As functions get more complicated, we need more clever techniques for antideriving/integrating.

Theorem 5.5.3 (The Substitution Rule (for Indefinite Integrals))

If $u = g(x)$ is a differentiable function whose range is an interval I, and f is continuous on I, then

$$\int f(g(x))g'(x)\,\mathrm{d}x = \int f(u)\,\mathrm{d}u.$$

Note: The Substitution Rule is the "inverse" of the Chain Rule (for derivatives).

Remark

There are two main things we look for when deciding to use a substitution, although there may be other less common applications.
- First, we use a substitution when we see funciton composition, or a function plugged into another function.
- Another use case is if the derivative (up to scaling by a constant) of part of the integrand also appears in the integrand.

Tip

To integrate $\int f(g(x))g'(x)\,\mathrm{d}x$:
1. Substitute $u = g(x)$ and $\mathrm{d}u = g'(x)\,\mathrm{d}x$
2. Integrate with respect to u.
3. Replace u with $g(x)$ in the result.

Example 5.5.4

Find

$$\int \frac{2x + 1}{x^2 + x + 3}\,\mathrm{d}x$$

Solution

We notice that $\frac{\mathrm{d}}{\mathrm{d}x}(x^2 + x + 3) = 2x + 1$. This is our indication to try a substitution:

$$\begin{aligned}\text{Let } u &= x^2 + x + 3\\ \frac{\mathrm{d}u}{\mathrm{d}x} &= 2x + 1\\ \mathrm{d}u &= 2x + 1\,\mathrm{d}x\end{aligned}$$

Substituting in u and $2x + 1\,\mathrm{d}x$ (note we could solve this for $\mathrm{d}x$ and plug that in instead) we get

$$\begin{aligned}\int \frac{2x+1}{x^2+x+3}\,\mathrm{d}x &= \int \frac{1}{u}\,\mathrm{d}u\\ &= \ln|u| + C\\ &= \ln|x^2+x+3| + C\end{aligned}$$

So far we have only discussed indefinite integrals, those without bound. For definite integrals we have to be careful about the bounds when we make the substitution, since the bounds are in terms of the original variable, not our substituted variable.

Theorem 5.5.5 (The Substitution Rule (for Definite Integrals))

If g' is continuous on $[a, b]$ and f is continuous on the range of $u = g(x)$, then

$$\int_a^b f(g(x))g'(x)\,\mathrm{d}x = \int_{g(a)}^{g(b)} f(u)\,\mathrm{d}u$$

♡

Warning

When computing a definite integral with a substitution, you have two options: switch the bounds of integration to be with respect to u **or** switch the variable of integration back to x after finding the antiderivative and use the original bounds. Note, if you do not switch the bounds, you cannot write the old bounds in terms of x in your work on steps where you are talking about u.

Both methods will be demonstrated in this section (and probably throughout these notes), you are free to choose your favorite method, use either, etc.

Example 5.5.6

Evaluate $\displaystyle\int_1^e \frac{(\ln(x))^2}{2x}\,\mathrm{d}x$.

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Solution

We see a function composition, $\ln(x)$ looks like it has been plugged into something else. We pick that as u.

$$\begin{aligned}\text{Let } u &= \ln(x)\\ \mathrm{d}u &= \frac{1}{x}\,\mathrm{d}x.\end{aligned}$$

We also need to change the bounds of integration

$$u(e) = \ln(e) = 1$$
$$u(1) = \ln(1) = 0$$

The integral then is

$$\int_1^e \frac{(\ln(x))^2}{2x}\,\mathrm{d}x = \int_0^1 \frac{u^2}{2}\,\mathrm{d}u$$
$$= \left[\frac{1}{6}u^3\right]_0^1$$
$$= \frac{1}{6}$$

Notice in the method above, since we changed the bounds of the integral to be in terms of u, we did not need to plug in $u = \ln(x)$ again. Changing the bounds tends to be faster in most examples, and if you don't change the bounds you have to be much more careful with notation.

Note

An optional bit of notation for clarity on bounds may be useful. To remember you have to change the bounds of integration when doing a substitution you can write the variable name in the lower bound. For example,

$$\int_{x=1}^e \frac{(\ln(x))^2}{2x}\,\mathrm{d}x = \int_{u=0}^1 \frac{u^2}{2}\,\mathrm{d}u.$$

This kind of extra notation for clarity can act as training wheels as you get used to the substitution step of changing bounds, but also this notation becomes *very* useful in multivariable calculus.

Here are some less obvious substitution problems.

Example 5.5.7

Evaluate $\displaystyle\int_0^{\frac{\pi}{16}} \tan(4x)\,\mathrm{d}x$.

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Solution

It seems that $4x$ has been plugged into the tangent function. However, using this as our u would not change the fact that we don't have a known antiderivative for $\tan(u)$. To see the substitution, sometimes we need to rewrite problems

$$\int_0^{\frac{\pi}{16}} \tan(4x)\,\mathrm{d}x = \int_0^{\frac{\pi}{16}} \frac{\sin(4x)}{\cos(4x)}\,\mathrm{d}x$$

Now we can see a function and (a constant multiple of) its derivative: $\frac{\mathrm{d}}{\mathrm{d}x}\cos(4x) = -4\sin(4x)$. This becomes our candidate for u

$$\begin{aligned}\text{Let } u &= \cos(4x)\\ \text{then } \mathrm{d}u &= -4\sin(4x)\,\mathrm{d}x\\ -\frac{1}{4}\,\mathrm{d}u &= \sin(4x)\,\mathrm{d}x\end{aligned}$$

(notice we can solve for $\mathrm{d}x$ and plug that in, but after we sub u we can see we have the $\sin(4x)\,\mathrm{d}x$ there anyway) and the bounds are changed to

$$\begin{aligned}u\left(\frac{\pi}{16}\right) &= \frac{\sqrt{2}}{2}\\ u(0) &= 1\end{aligned}$$

then

$$\int_0^{\frac{\pi}{16}} \frac{\sin(4x)}{\cos(4x)}\,\mathrm{d}x = \int_1^{\frac{\sqrt{2}}{2}} -\frac{1}{4}\frac{1}{u}\,\mathrm{d}u.$$

The antiderivative of $\frac{1}{u}$ is $\ln(u)$. So,

$$\begin{aligned}\int_0^{\frac{\pi}{16}} \frac{\sin(4x)}{\cos(4x)}\,\mathrm{d}x &= \left[-\frac{1}{4}\ln|u|\right]_1^{\frac{\sqrt{2}}{2}}\\ &= -\frac{1}{4}\ln\left(\frac{\sqrt{2}}{2}\right)\end{aligned}$$

(Optionally,) This could be further simplified by $-\frac{1}{4}\ln\left(\frac{\sqrt{2}}{2}\right) = -\frac{1}{4}\left(\ln\left(\sqrt{2}-\ln(2)\right)\right) = -\frac{1}{4}\left(\frac{1}{2}\ln(2)-\ln(2)\right) = -\frac{1}{4}\left(-\frac{1}{2}\ln(2)\right) = \frac{\ln(2)}{8}$

Example 5.5.8

Find $\displaystyle\int x^2\sqrt[3]{x+1}\,\mathrm{d}x$

◇

This is another common type of substitution problem that doesn't follow the usual patterns. You can recognize these types of problems by the product of a polynomial (in this case x^2) and another polynomial to some exponent (in this case $\sqrt[3]{x+1}$).

Solution

Let $u = x+1$. Then $\mathrm{d}u = \mathrm{d}x$. Note, that we also need something to plug into the x^2 term, so solving for x we have $x = u-1$ so $x^2 = (u-1)^2$. Then,

$$\begin{aligned}\int x^2\sqrt[3]{x+1}\,\mathrm{d}x &= \int (u-1)^2\sqrt[3]{u}\,\mathrm{d}u\\ &= \int (u^2-2u+1)\sqrt[3]{u}\,\mathrm{d}u\\ &= \int \left(u^2u^{\frac{1}{3}} - 2uu^{\frac{1}{3}} + 1u^{\frac{1}{3}}\right)\mathrm{d}u\\ &= \int u^{\frac{7}{3}} - 2u^{\frac{4}{3}} + u^{\frac{1}{3}}\,\mathrm{d}u\end{aligned}$$

Note: because we moved the polynomial with multiple terms to the integer exponent, we could manually expand and distribute. Now we only need to use power rule a few times. Integrating we get

$$\begin{aligned}\int u^{\frac{7}{3}} - 2u^{\frac{4}{3}} + u^{\frac{1}{3}}\,\mathrm{d}u &= \frac{3}{10}u^{\frac{10}{3}} - \frac{6}{7}u^{\frac{7}{3}} + \frac{3}{4}u^{\frac{4}{3}} + C \\ &= \frac{3}{10}(x+1)^{\frac{10}{3}} - \frac{6}{7}(x+1)^{\frac{7}{3}} + \frac{3}{4}(x+1)^{\frac{4}{3}} + C.\end{aligned}$$

5.5 Section Summary:

- We reviewed the fundamental theorem of calculus, linking the area under a curve to the antiderivative of the function.
- We learned about one solution method for integrals: making a substitution.
- We noticed how the bounds of the integral change when we make substitutions.

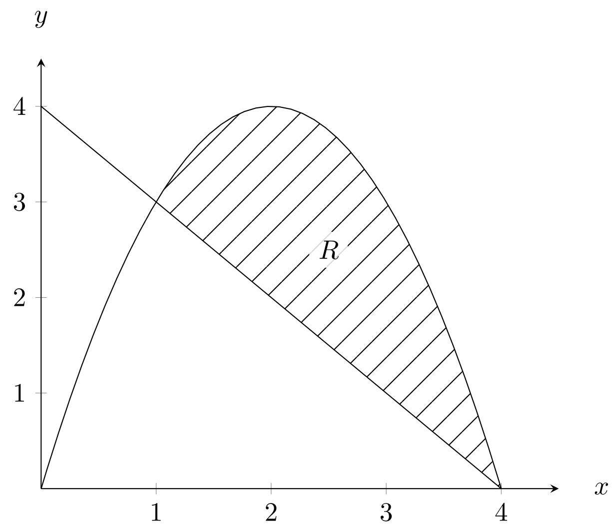
6. A Few Integral Applications

6.1. Area Between Curves

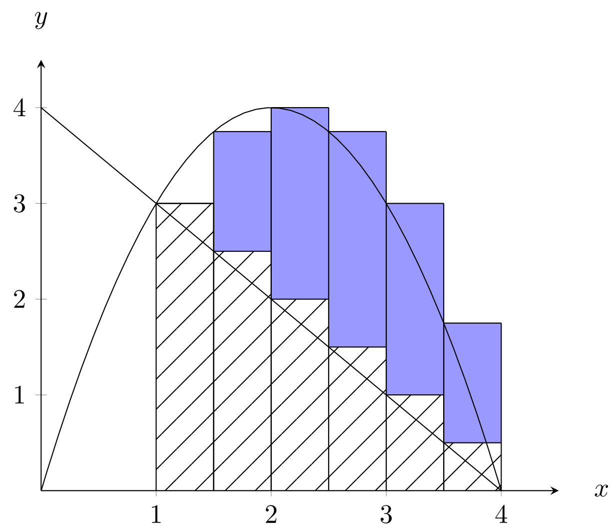
6.1 Learning Objectives
- I can extend the idea of area under a curve to area between curves.

Suppose we wish to find the area between the curves $y = -x^2 + 4x$ and $y = -x + 4$ on $[1, 4]$.

How can we approximate this area?



To approximate this area, we can draw rectangles between the curves, with bottom edge lined up with the lower curve, and top edge lined up with the upper curve. This is effectively the same as estimating the area under each curve, and taking the difference between them (with the same number of rectangles). Note in the following plot, the first blue rectangle is the same height as the dashed one (because the functions intersect at the left side of this sub-interval), so it is not diplayed.



We can approximate the area under both curves individually. Then, since the area under the “lower” curve is under both curves, we can subtract it away to get the area that is left.

Definition 6.1.1 (Area Between Curves)

Let R denote the region between continuous functions $y = f(x)$ and $y = g(x)$ and between the vertical lines $x = a$ and $x = b$, given that $f(x) \geq g(x)$ on $[a, b]$. The area of R is defined by

$$\begin{aligned} A &= \left(\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x\right) - \left(\lim_{n\to\infty} \sum_{i=1}^{n} g(x_i^*)\Delta x\right) \\ &= \lim_{n\to\infty} \sum_{i=1}^{n} (f(x_i^*) - g(x_i^*))\Delta x. \end{aligned}$$

where we divide R into n strips of equal width, $\Delta x = \frac{b-a}{n}$ and height $f(x_i^*) - g(x_i^*)$. Thus, written as a definite integral, $A = \int_a^b (f(x) - g(x))\, \mathrm{d}x$.

Note

It isn't really necessary to define the area between curves in terms of Reimann sums. So long as you can reason out why the area between curves is the integral of their difference, you are good to go. You do also need to know the limit of Reimann sums definition of the integral, so it is good practice to go over it again here in a slightly more complicated format. Discuss in office hours if you have questions about this formulation.

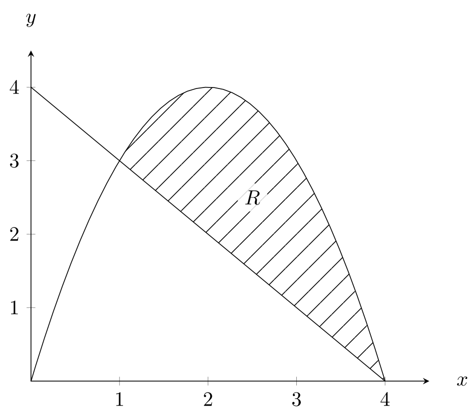
Exercise 6.1.2

Find the area between the curves $y = -x^2 + 4x$ and $y = -x + 4$ on $[1, 4]$.

Solution

- It is good to sketch the region in these cases, and find any intersection points. Remember that we only get the area between curves if the upper function is, well, the upper function (we require $f(x) \geq g(x)$ on $[a, b]$ in the area between curves definition). If the functions changed which one was greater than the other over our interval, we would just break up the interval and compute a few integrals.

The region R was graphed before but is reproduced here for completion:



Note, on $[1, 4]$ we always have $-x^2 + 4x \geq -x + 4$. Then, the area, A, between curves is

$$\begin{aligned}
A &= \int_1^4 (-x^2 + 4x) - (-x + 4)\,\mathrm{d}x \\
&= \int_1^4 -x^2 + 5x - 4\,\mathrm{d}x \\
&= \left[-\frac{1}{3}x^3 + \frac{5}{2}x^2 - 4x\right]_1^4 \\
&= -\frac{1}{3}(4)^3 + \frac{5}{2}(4)^2 - 4(4) - \left(-\frac{1}{3}(1)^3 + \frac{5}{2}(1)^2 - 4(1)\right) \\
&= \frac{19}{2}
\end{aligned}$$

Exercise 6.1.3

1. Compute the are between the curves $y = -x^2 + 4x$ and $y = -x + 4$ on $[0, 2]$. *Note: you will need to compute this using 2 definite integrals, make sure to check that the upper function is the first one in your difference.*
2. Convince yourself that the area between curves definition works even when $g(x) \leq 0$. You should be finding the *unsigned* area when you do this, what does that tell you about the sign of your answer to a problem asking for the area between curves?

Instead of taking vertical rectangles to approximate the area between curves, we might look at horizontal rectangles to find area between curves. This is especially useful in cases where we have more of a “right” and “left” curve instead of an “upper” and “lower” curve, as in the following example.

Example 6.1.4

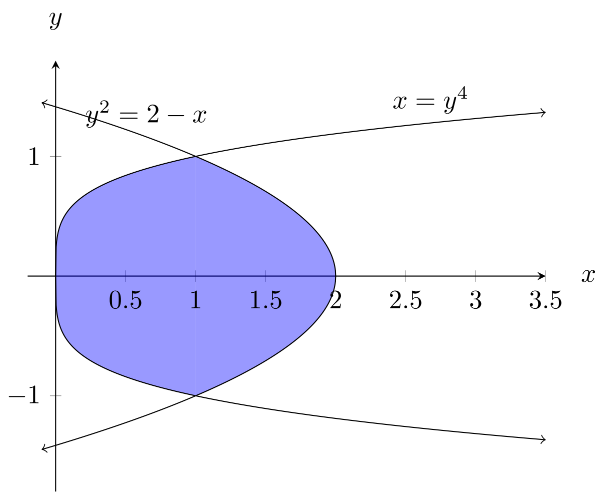
Set up and evaluate an integral to find the area enclosed by $x = y^4$ and $y^2 = 2 - x$.

Warning

It is expected that you know how to graph basic functions and similar relations like the ones above, although you may not have been formally trained in a class on these specifically. Please come discuss this in office hours if the hint to “turn your head sideways and think about swapping x and y” does not help you. It is perfectly reasonably to be a little stuck on this when you first see it, and that is what office hours are there for!

Solution

First, we sketch the region to get an idea of what it looks like.



We see we have more of a right curve ($y^2 = 2 - x$) and left curve ($x = y^4$). In terms of x values, we have $x = y^4$ and $x = 2 - y^2$.

We start by finding where the functions intersect

$$\begin{aligned} y^4 &= 2 - y^2 \\ y^4 + y^2 - 2 &= 0 \\ (y^2 + 2)(y^2 - 1) &= 0 \end{aligned}$$

since $y^2 + 2$ is never 0, we know they intersect when $y^2 - 1 = 0$. To find the area between them, we can think about turning our head sideways and doing an integral in y, where instead of "upper function minus lower function" in the integral, we do "right function minus left function" (since $2 - y^2 \leq y^4$ for y on $[-1, 1]$ our method still applies). Thinking of this as an integral in y we have

$$\begin{aligned} \int_{-1}^{1} (2 - y^2) - y^4 \, \mathrm{d}y &= \left[2y - \frac{1}{3}y^3 - \frac{1}{5}y^5\right]_{-1}^{1} \\ &= 2 - \frac{1}{3} - \frac{1}{5} - \left(-2 + \frac{1}{3} + \frac{1}{5}\right) \\ &= \frac{44}{15} \end{aligned}$$

Example 6.1.5

Set up an integral to find the area enclosed between $f(x) = \cos(x)$ and $g(x) = \sin(x)$ on $[0, 2\pi]$.

◇

Solution

Sketching the graphs on $[0, 2\pi]$ and using our knowledge of the unit circle, we know that the functions intersect at $x = \frac{\pi}{4}, \frac{5\pi}{4}$. For $[0, \frac{\pi}{4}]$, $f(x) \geq g(x)$ and then on $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$, $g(x) \geq f(x)$, then at last on $\left[\frac{5\pi}{4}, 2\pi\right]$, $f(x) \geq g(x)$ once again. So, the area between the curves, A, is given by

$$A = \int_0^{\frac{\pi}{4}} (\cos(x) - \sin(x)) \, \mathrm{d}x + \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin(x) - \cos(x)) \, \mathrm{d}x + \int_{\frac{5\pi}{4}}^{2\pi} (\cos(x) - \sin(x)) \, \mathrm{d}x$$

Tip

To find the area between 2 curves do the following

1. Identify the domain, so you have a good idea of what the region is.
2. Find where the curves intersect: set the equations of the curves equal to each other, and make sure to consider any domain restrictions.
3. Determine the subintervals over which you need to calculate the area: does the "upper curve" switch over the interval?
4. Determine whether $f(x) \geq g(x)$ or $g(x) \geq f(x)$ on each sub-interval.
5. Set up the integrals for each sub-interval with the greater function minus the smaller function ("upper minus lower" or "right minus left") and evaluate.
6. Check: We are looking for *unsigned* area when we ask for the area between the curves, you should always get a positive number!

Exercise 6.1.6

Set up an integral to find the area of the triangular region enclosed by the points $(0, 0), (2, 2), (5, 0)$ with respect to both x and y.

◇

6.1 Section Summary:

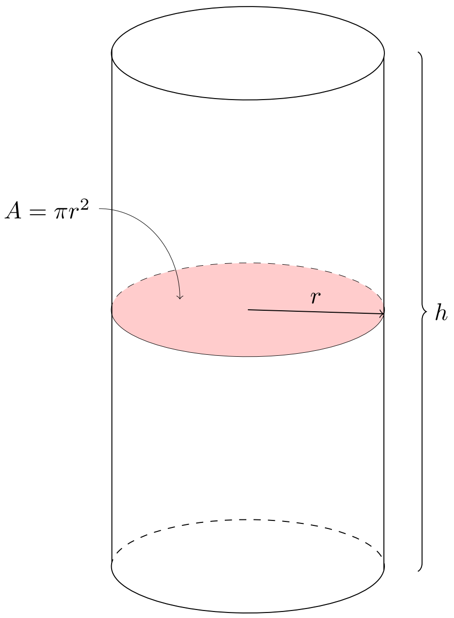
- We learned how to find the area between curves, either in the x or y direction.

6.2. Volumes by Cross-Sections, Disks, and Washers

6.2 Learning Objectives:

- I can think of *cross-sectional area* when calculating the volume of certain shapes using definite integrals.
- I can use disk or washer method to find the volume of a *solid of revolution*. I can recognize that these methods are related to finding the area between curves.

Recall how we find the volume of a cylinder. $V = \pi r^2 h$. We essentially get the volume by taking the area of the base, πr^2 and sweeping it along the entire height h.

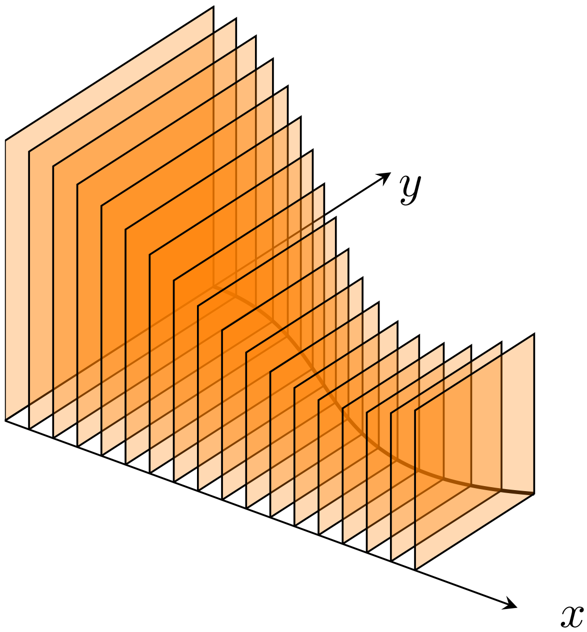


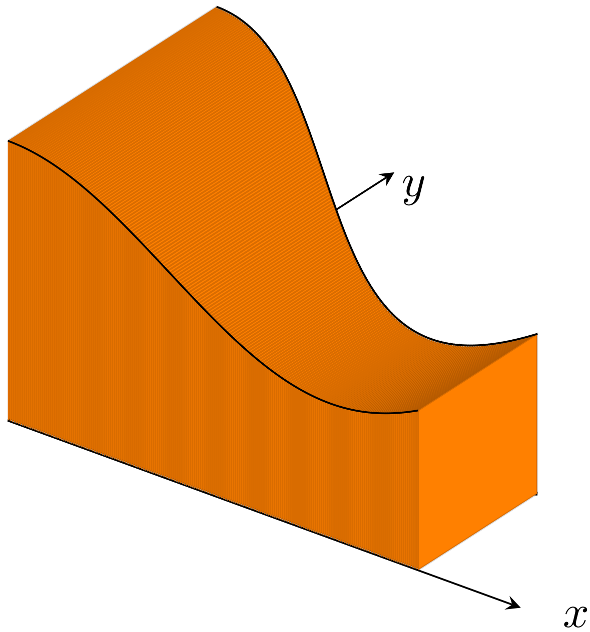
6.2.1. Slicing by Parallel Planes

Definition 6.2.1

We define a *cross-section* of a solid S as the plane region formed by intersecting S with a plane. ♣

We can build a (3 dimensional) solid S by taking the same type of cross-section (e.g. squares) for each value of x, based on some region in the Cartesian plane.



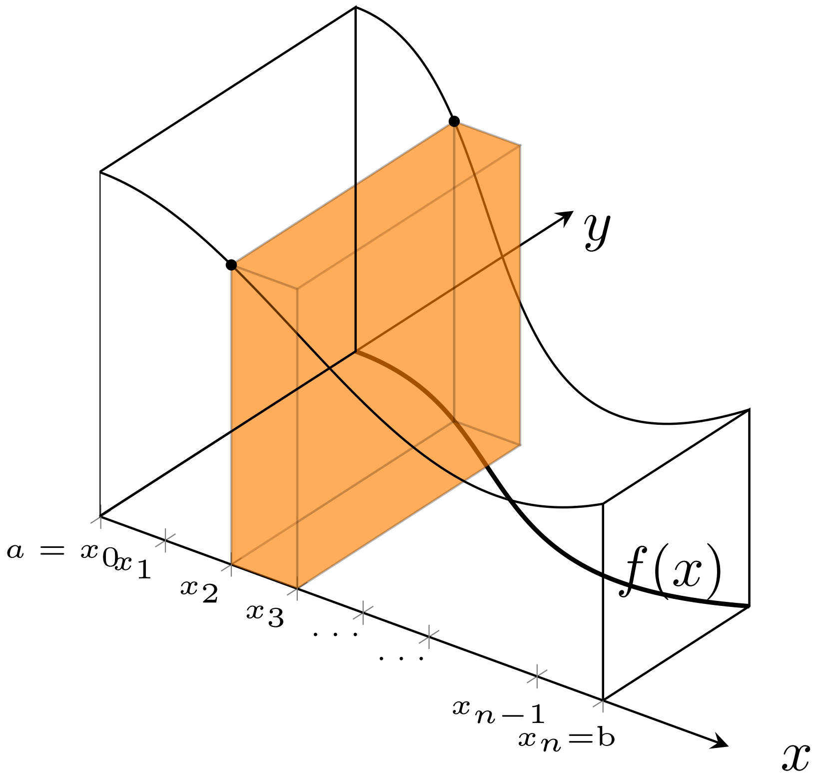


Suppose that we partition $[a, b]$ into subintervals

$$a = x_0 < x_1 < x_2 < x_3 < \ldots < x_n = b$$

where each subinterval has width Δx. What is the volume of one slice of our solid? How would we use this to calculate the voluem of the entire solid?

To approximate the volume, we can think of a cross-section with width Δx, then its volume is $V = (\text{Area of section})\Delta x$. Here the area of the cross-section is the area of a square with side length equal to the function value, $y = f(x)$.



As the width of the sub-intervals goes to zero and we add more rectangular prisms together, we get a better approximation of the volume. Once again, the limit of this sum ends up being a definite integral.

Definition 6.2.2

If the cross-section of a solid S at each point x in $[a, b]$ is a region $S(x)$ of area $A(x)$, where $A(x)$ is a continuous function of x, then

$$V = \int_a^b A(x)\,\mathrm{d}x.$$

♣

Tip

To calculate the volume of a solid

1. Sketch a picture of the region and a typical cross-section.
   - Don't overcomplicate the cross-section, it should just be a prism with straight sides and top/bottom (or left/right sides) the same shape with area $A(x)$ (the area changes for each cross-section but is the same on the 2 faces of the single cross-section). It is never a part of a cone or pyramid or sphere,,,
2. Find a formula for $A(x)$.
3. Find the limits of integration.

- Think about stretching the face with area $A(x)$ from one end of the shape to the other, where do you start/stop this process?

4. Integrate to find the volume, mind your units if applicable.

Note

Typical cross-sectional area formulae:

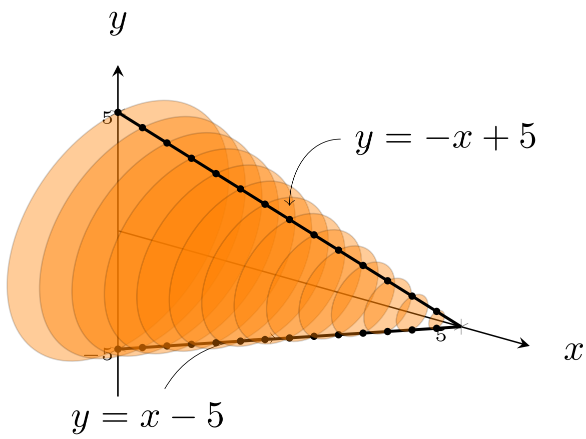
- Square: for a square with side length s, the area is $A = s^2$.
- Triangle: for a triangle with base b and height h, the area is $A = \frac{1}{2}bh$.
- Circle: for a circle with radius r, the area is $A = \pi r^2$.
- Semicircle: for a circle with radius r, the area is $A = \frac{1}{2}\pi r^2$.

Example 6.2.3

A cone with height 5 cm has a circular base with radius 5 cm. Set up a definite integral and evaluate it to find the volume of the cone.

Solution

A sketch of the volume, showing cross-sections, might look like:



We can think of slices of the cone being circles, with radius equal to the y value in $y = -x + 5$. This comes from the edge of the cone making the line $y = -x + 5$ in the x-y plane.

Then the area of one slice is $A(x) = \pi(-x + 5)^2$. The slices range from $x = 0$ to $x = 5$.

The definite integral we need to evaluate is

$$\begin{aligned} V &= \int_0^5 \pi(-x+5)^2 \, \mathrm{d}x \\ &= \pi \int_0^5 x^2 - 10x + 25 \, \mathrm{d}x \\ &= \pi \left[\frac{1}{3}x^3 - 5x^2 + 25x\right]_0^5 \\ &= \pi \left[\frac{1}{3}(5)^3 - 5(5)^2 + 25(5)\right] \\ &= \frac{125}{3}\pi \end{aligned}$$

We are told the distances are measured in centimeters, so the final answer is the volume is $\frac{125\pi}{3}$ centimeters cubed.

There are 2 checks we can do at the end of the problem. First, our solution is positive, which we expect because volume is positive. Second, this is a cone and we do know the volume formula for a cone:

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(5)^2(5) = \frac{125}{3}\pi.$$

This is the same thing we got from integrating, so our answer is correct!

Exercise 6.2.4

Derive the formula for the volume of a cone using similar techniques as above. *Hint: the line connecting the points* $(0, r)$ *and* $(h, 0)$ *will be useful.*

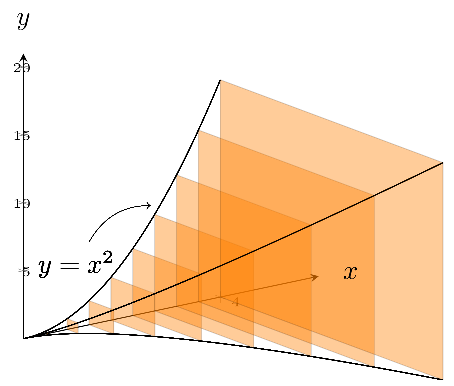
Example 6.2.5

Set up an integral to find the volume of the given solid. The base of the solid is the region bounded by $x = 0, x = 4$, and $y = x^2$. The cross-sections are perpendicular to the x-axis and are squares.

Set up an integral to find the volume of the given solid with the same base and square cross-sections, but the cross-sections are perpendicular to the y-axis.

Solution

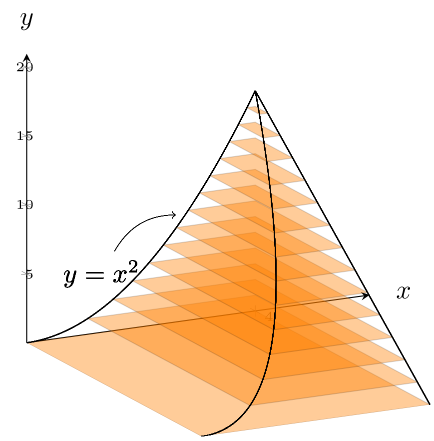
Sketch the region first, then



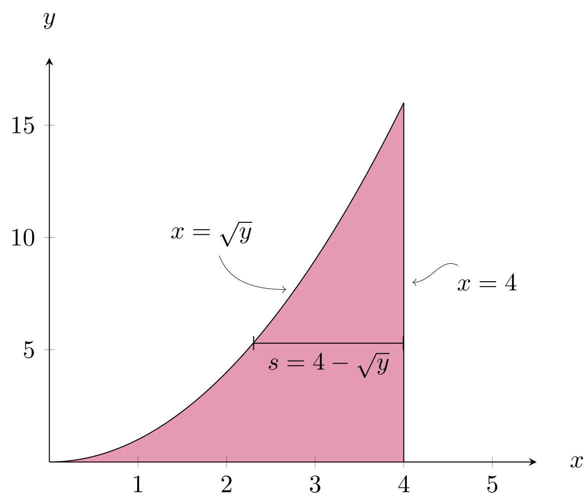
Where we see the side length of the square cross-sections is x^2, and the slices fill the space from $x = 0$ to $x = 4$ so the definite integral we need to evaluate for the volume is

$$V = \int_0^4 \left(x^2\right)^2 \, \mathrm{d}x = \int_0^4 x^4 \, \mathrm{d}x.$$

For the second part of the problem, we are now looking at the region below:



Notice that the slices now go from bottom to top, when the cross-sections are perpendicular to y-axis, we want to use an integral in y. This region has square cross-sections where we can think about "right curve minus left curve" to find the side length of those squares:



Here the square side stretches between $x = \sqrt{y}$ and the vertical line $x = 4$. So, the side length is $s = 4 - \sqrt{y}$. This solid is formed up of slices from $y = 0$ to $y = 16$. Then the integral we solve for the volume is

$$V = \int_0^{16} (4 - \sqrt{y})^2 \, dx.$$

Exercise 6.2.6

Predict if you think the volumes of the two solids above will be the same or different. Then, evaluate each integral and compare. Were you correct or incorrect in your prediction? If you were incorrect, that is okay, but try to identify why/what went wrong.

Remark

In these problems, the variable of integration is determined based on whether the cross-sections are perpendicular to the x-axis or y-axis.

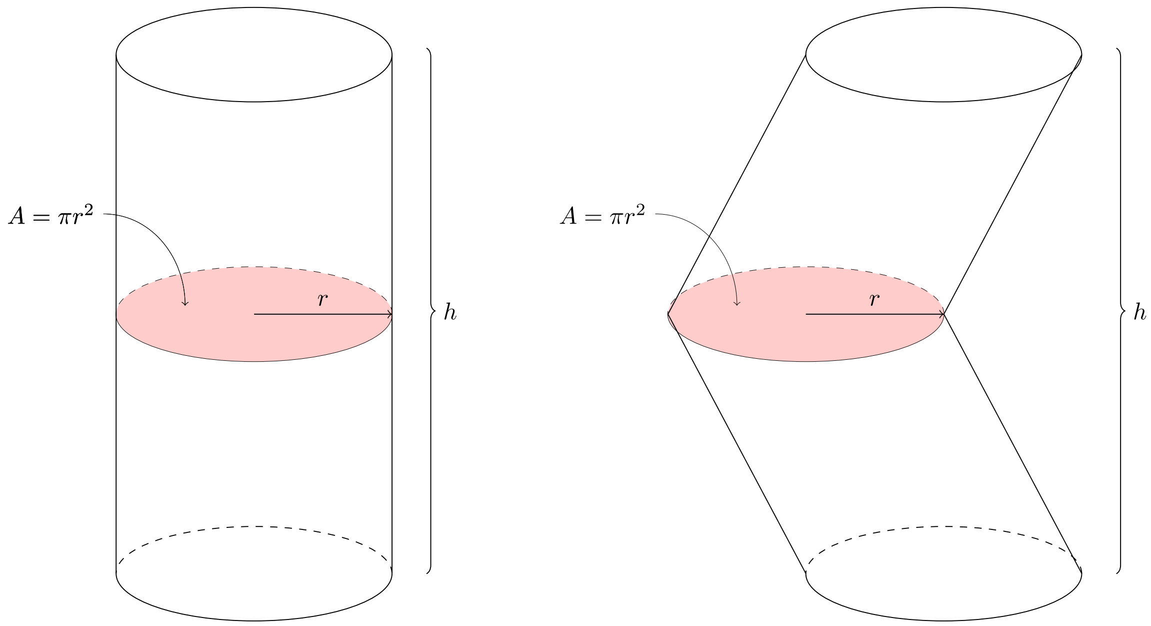
- Slices perpendicular to the x-axis (where slices stack from one x value to another) mean the variable of integration is x.
- Slices perpendicular to the y-axis (where slices stack from one y value to another) mean the variable of integration is y.

Property 6.2.7 (Cavalieri's Principle)

Solids with equal *height* and identical *cross-sectional area* at each height have the same volume.

Remark

These 2 cylinder-like solids have the same volume by Cavalieri's Principle



Exercise 6.2.8

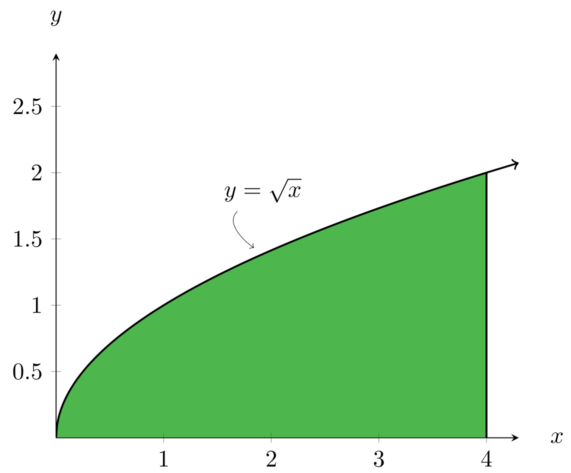
Set up an integral to find the volume of the given solid. The base of the solid is the region bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{2}$. The cross-sections perpendicular to the x-axis are equilateral triangles. ♢

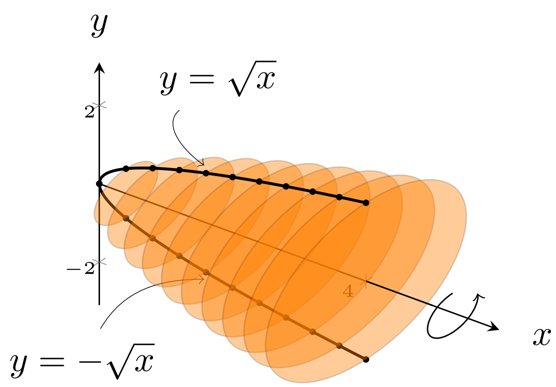
6.2.2. Solids of Revolution: The Disk method

Definition 6.2.9

A *solid of revolution* is the solid generated by rotating (or revolving) a plane region about an axis in its plane. ♣

Consider the function $y = \sqrt{x}$ on $[0, 4]$. Imagine rotating it all the way around the x-axis to form a solid:





Note

If we wish to find the volume of this solid, we have only added one step to our process from before. After that we have the same type of problem as the ones with the cross-sections. We have to build the 2-D region we want to form the solid from.

1. Sketch the function and its reflection about the axis you are told to revolve around. This with the domain give you the 2-D region we go from.
2. Imagine circular cross-sections that are perpendicular to the same axis you just used.
3. Continuing as before the cross-sections will *always* be circular and have radius equal to the function value.

We can make this more specific below.

Theorem 6.2.10 (Volume by Disks for Rotation about the x-axis)

To find the volume of a solid formed by rotating a function $y = f(x)$ on $[a, b]$ about the x-axis first note that the area of each cross-section is $A(x) = \pi(f(x))^2$, then the volume is

$$V = \int_a^b A(x)\,\mathrm{d}x = \int_a^b \pi(f(x))^2\,\mathrm{d}x.$$

♡

Exercise 6.2.11

Write a similar statement to the one above for rotating a function $y = f(x)$ for y values in $[a, b]$ about the y-axis. *Hint: careful with the variables you use in the expression. The variable of integration should be the variable used in the integrand.*

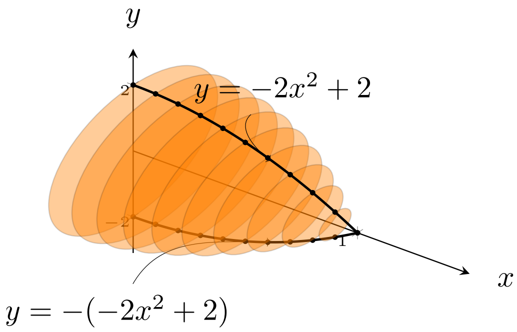
Example 6.2.12

Set up an integral to find the volume of the solid generated by revolving the region in the first quadrant bounded above by $y = -2x^2 + 2$ around the x-axis.

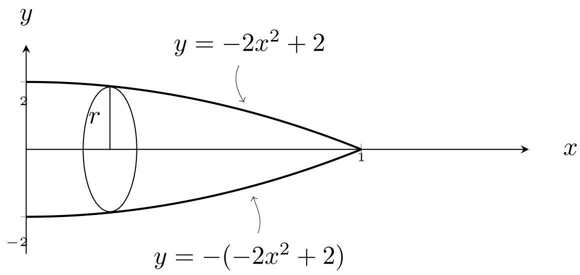
Set up an integral to find the volume when revolving this same region around the y-axis.

Solution

The solid looks like



However, this might be difficult to draw by hand. An alternative is to just draw the curve in 2-D, its reflection, and a single cross-section to help visualize the radius of the disks. This looks like



Then, we need to find where the curve hits the x-axis. Sometimes we can tell from the graph we draw, sometimes we have to solve for it

$$\begin{aligned}
-2x^2 + 2 &= 0 \\
-2(x^2 - 1) &= 0 \\
-2(x+1)(x-1) &= 0 \\
x &= 1, -1
\end{aligned}$$

Since we are looking in the first quadrant, $x = 1$ is where this intersection happens (This is one of the bounds for the integral later). The area of the circular cross-section is $\pi r^2 = \pi(-2x^2 + 2)^2$. Thus, the integral we need is $V = \int_0^1 \pi(-2x^2 + 2)^2 \, dx$.

Exercise 6.2.13

Evaluate the integral above to find the volume of the solid.

Exercise 6.2.14

Set up and solve an integral to find the volume when revolving this same region around the y-axis.

Theorem 6.2.15 (Volume by Disks for Rotation about the y-axis)

To find the volume of a solid formed by rotating a function $y = f(x)$ (and solving for $x = f^{-1}(y)$) for x in $[a, b]$ and y between $f(a)$ and $f(b)$ about the y-axis first note that the area of each cross-section is $A(y) = \pi(f^{-1}(y))^2$, then the volume is

$$V = \int_{f(a)}^{f(b)} A(y) \, dy = \int_{f(a)}^{f(b)} \pi(f^{-1}(y))^2 \, dy.$$

In other words,

Note

Finding the Volume of a Solid of Revolution: Disk Method

1. Sketch both the region that is being rotated and the axis of revolution.
2. Determine the variable of integration. Integration is performed with respect to the axis that is perpendicular to the slices.
3. Determine the limits of integration. They must match the variable of integration!
4. Derive a formula for the radius of the disk.
5. Integrate. Remember units in your final answer when applicable.

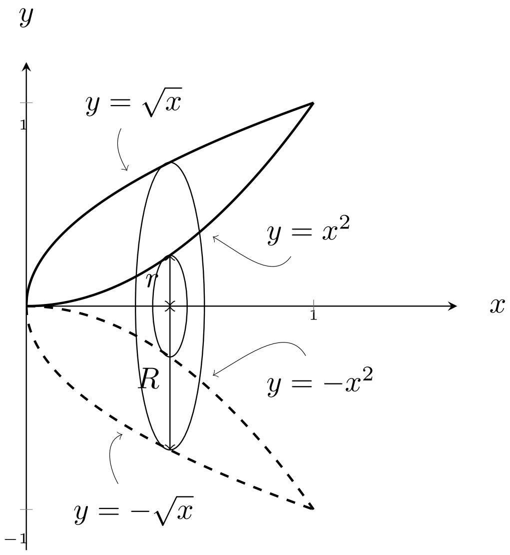
6.2.3. Solids of Revolution: The Washer Method

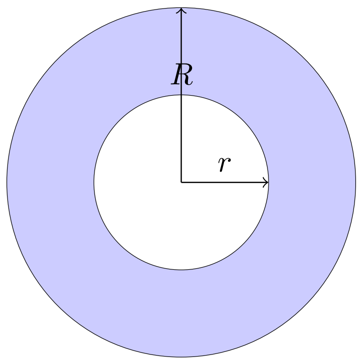
Note that we can only use the disk method if the cross-sections of the solid of revolution are circles. This means that there is no space gap between the region and the axis of revolution. What if there is a gap between the region and the axis of revolution?

Example 6.2.16

Set up an integral to find the volume of the solid generated by revolving the region in the first quadrant bounded by $y = x^2$ and $y = \sqrt{x}$ around the x-axis.

Solution





We note the curves intersect at $(1, 1)$. There are a few ways to think about this problem, but for now let's go back to thinking about the area of the cross-section and integrating that from 0 to 1 in x. For the cross-sections, the area is a circle with radius $R = \sqrt{x}$ minus the area of a smaller circle with area $r = x^2$. This region is sketched above in blue, and is known as a washer (as in the metal bit used with a nut and bolt that has the same shape). The area of the full region is the difference in areas between the 2 circles. That is, $A(x) = \pi R(x)^2 - \pi r(x)^2$. Then, integrating along the axis perpendicular to the cross-sectional area, we see the volume is

$$
\begin{aligned}
V &= \int_0^1 A(x)\,\mathrm{d}x \\
&= \int_0^1 \pi R(x)^2 - \pi r(x)^2\,\mathrm{d}x \\
&= \pi \int_0^1 \left[\left(\sqrt{x}\right)^2 - \left(x^2\right)^2\right] \mathrm{d}x \\
&= \pi \int_0^1 x - x^4\,\mathrm{d}x
\end{aligned}
$$

Exercise 6.2.17

Evaluate the integral above. ♢

Theorem 6.2.18 (Volume by Washers for Rotation about the x-axis)

For a solid of rotation formed by rotating a region around the x-axis where you have the required bounds and radii in terms of x, the volume is

$$V = \int_a^b A(x)\,\mathrm{d}x = \pi \int_a^b \left[(R(x))^2 - (r(x))^2\right] \mathrm{d}x$$

Theorem 6.2.19 (Volume by Washers for Rotation about the y-axis)

For a solid of rotation formed by rotating a region around the y-axis where you have the required bounds and radii in terms of y, the volume is

$$V = \int_a^b A(y)\,\mathrm{d}y = \pi \int_a^b \left[(R(y))^2 - (r(y))^2\right] \mathrm{d}y$$

In other words,

Note

Finding the Volume of a Solid of Revolution: Washer Method

1. Sketch both the region that is being rotated and the axis of revolution.
2. Determine the variable of integration. Integration is performed with respect to the axis that is perpendicular to the slices.
3. Determine the limits of integration. They must match the variable of integration!
4. Derive a formula for the radii of the washer: $R(x)$ and $r(x)$ (or $R(y)$ and $r(y)$ respectively).
   - These are found by finding the distances of each curve from the axis of revolution. For more, see area between curves section.
5. Integrate. Remember units in your final answer when applicable.

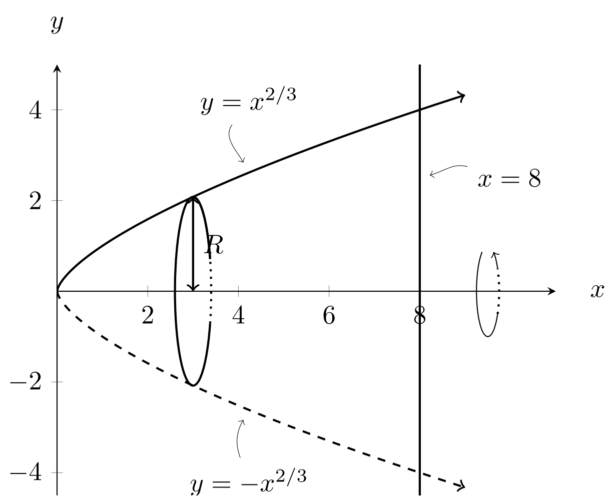
Example 6.2.20

Set up an integral to find the volume of the solid generated by revolving the region bounded by $y = x^{\frac{2}{3}}$, the x-axis, and $x = 8$

a) around the x-axis
b) around the y-axis
c) around the line $y = 6$.

Solution

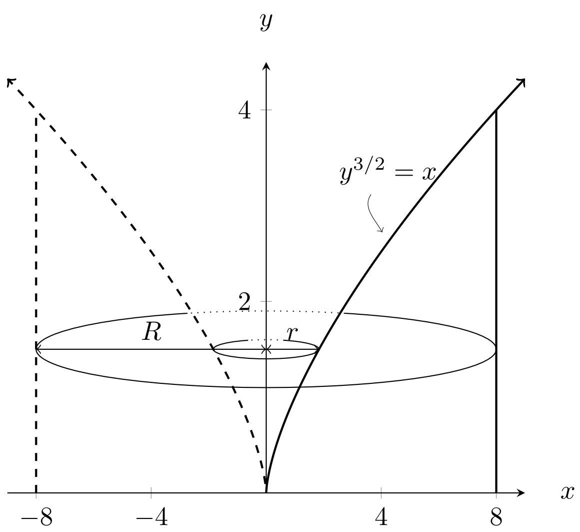
a) Sketching the region and an idea of the volume we make by rotating around the x-axis we see the cross-sections are disks.



Thus, the volume can be found with the disk method: by thinking of the cross-sectional area as a bunch of circles perpendicular to the x-axis, from $x = 0$ to $x = 8$.

$$\begin{aligned} V &= \pi \int_0^8 (R(x))^2 \,\mathrm{d}x \\ &= \pi \int_0^8 \left(x^{\frac{2}{3}}\right)^2 \mathrm{d}x \\ &= \pi \int_0^8 x^{\frac{4}{3}} \,\mathrm{d}x \end{aligned}$$

a) Sketching the region and an idea of the volume we make by rotating around the y-axis we see the cross-sections are washers now.



Notice that we need to express the radii in terms of y now, the large radius is always $R(y) = 8$ and the small radius is $r(y) = y^{\frac{3}{2}}$ which we got from solving the original equation for x. Lastly, we note the solid fills space for y in $[0, 4]$. Then the volume integral will be

$$\begin{aligned} V &= \int_0^4 \pi\left[(R(y))^2 - (r(y))^2\right] \mathrm{d}y \\ &= \int_0^4 \pi\left[(8)^2 - \left(y^{\frac{3}{2}}\right)^2\right] \mathrm{d}y \\ &= \pi \int_0^4 \left[64 - y^3\right] \mathrm{d}y \end{aligned}$$

a) Sketching the region and the horizontal line $y = 6$ we have

6.2 Section Summary:

6.3. Volumes by Cylindrical Shells

6.3 Learning Objectives:

6.3 Section Summary:

6.4. Work

6.4 Learning Objectives:

6.4 Section Summary: